

MTConnect[®] Institute

<https://projects.mtconnect.org/account/register>

- Be part of a growing community.
- Provide valuable insight of your use cases for inclusion in the standard.
- Contribute to its development.

Resources

<https://www.mtconnect.org/>

<https://www.mtcup.org/>

<https://projects.mtconnect.org/>

<https://github.com/mtconnect>

<https://www.youtube.com/user/MTConnect>



MTCONNECT
VERSION 2.0
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Session 6

Validation of MTConnect Output

Helpful Links

SysML Model Browser – <https://model.mtconnect.org/>

Navigate the MTConnect Standard model on the web.

GitHub Repo – https://github.com/mtconnect/mtconnect_sysml_model

Access the MTConnect Standard model on GitHub.

TAMS Validator – <https://validator.tams.ai/>

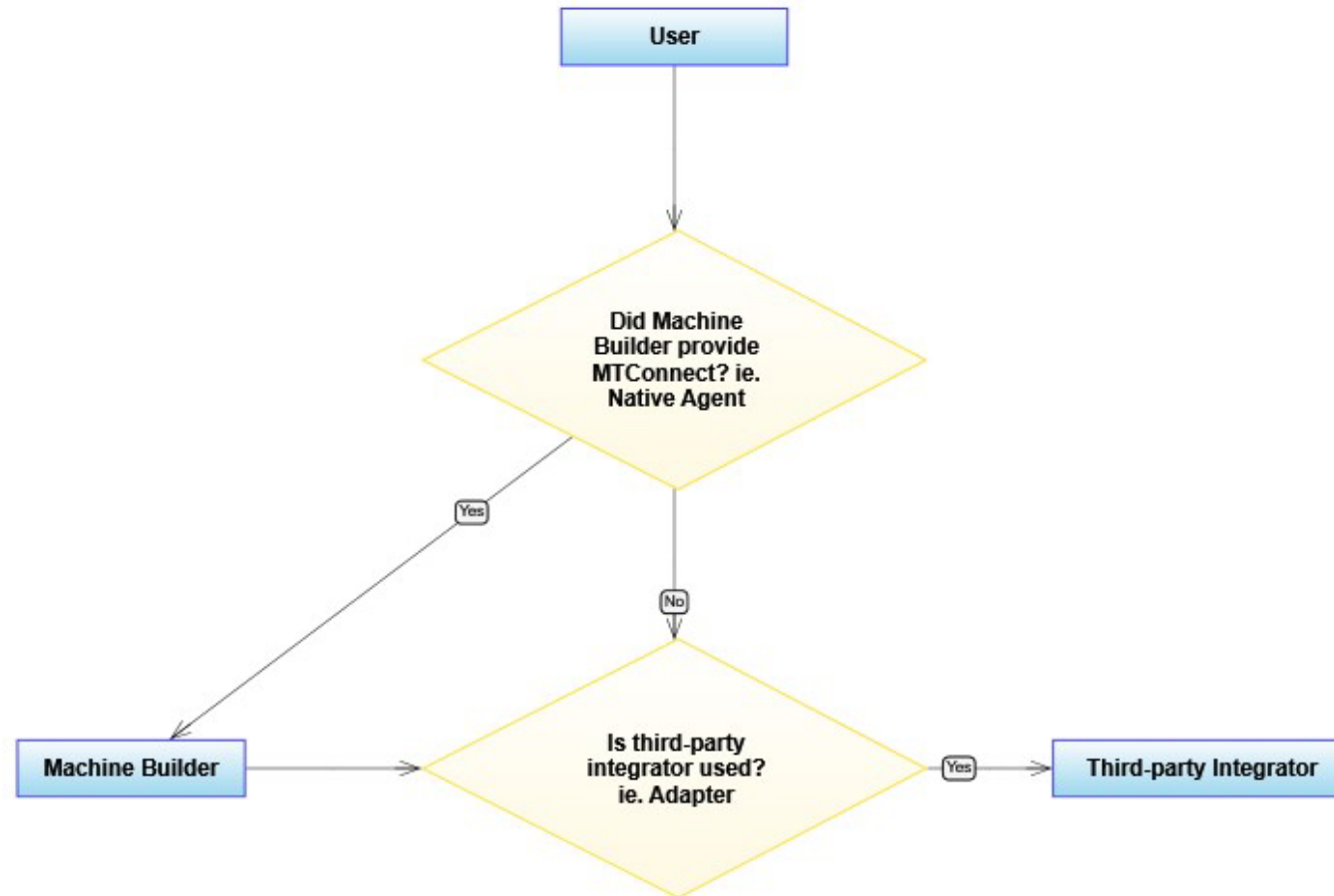
Free spot-check validation of Device models.

Actors and Domain Knowledge Requirements

- **Implementers** (publisher, e.g. Builder, Third-Party Integrator)
- **Application** (subscriber, e.g. Monitoring Platform)
- **End User** (observer, e.g. Shop Manager)

Requirements	Implementer	Application	End User
Equipment data acquisition	Substantial	None	None
MTConnect Vocabulary	Substantial	Moderate	Minimal
MTConnect Semantics	Substantial	Moderate	Minimal
MTConnect Protocol	Substantial	Substantial	None

Who's Responsible for Adherence?



What is Validation?

Compliance to the constraints defined in MTConnect.

Structural

A degree of adherence that measures how well a MTConnect Response Document follows the structural constraints of the standard. For example, the provided observational type is defined in the MTConnect Standard.

Behavioral

A degree of adherence that measures how well a MTConnect Response Document is accessed according to the guidelines described in the MTConnect Standard. For example, does a MTConnect Agent implementation provide REST endpoints for probe, current, and sample with the appropriate parameters defined in the MTConnect Standard?

Capabilities

The concept that validation of a MTConnect implementation should be performed progressively starting with Structural Adherence, then Behavioral Adherence, and finally Capabilities Adherence. The validation of subsequent degrees is moot if the previous degrees do not pass.

When should I Validate?

- Purchase Order Requirements
 - E.g. New equipment, retrofits, third-party integrators
- Before Persistent Storage
 - Avoid “Garbage In – Garbage Out”
 - Applies to Databases, Data Lakes, Excel spreadsheets, etc.
- Realtime Dashboards
 - Use “Active Validation”, if available

Where does Validation fit?

- Regulated Process Data
- Data Science
 - E.g. Production Team led monitoring, but passes analytics to Data Scientists
 - E.g. Company implements AI utilizing MTConnect's Unified Information Model
- Supply Chain data propagation
 - E.g. Tier 1 supplier requires process data with QIF documents for ITAR/CMMC

Why Should I Validate MTConnect?

- Plug-and-Play Client Software
- Artificial Intelligence
- Mix of equipment/sensors
 - E.g. Types, Makes, Models, Vintages
- Multiple Facilities or Remote Monitoring
- Regulated Process Data

Validating Versions

Important Concepts

- Things get deprecated across versions
- Respect the context of the intended version for Structural and Behavioral validation

For example:

```
<Header version="1.3" ... />  
...  
<Execution>PROGRAM_OPTIONAL_STOP</Execution>
```

PROGRAM_OPTIONAL_STOP wasn't deprecated until v1.4, so this is valid

How do you validate MTConnect Structure?

- Get a MTConnect Response Document from an Agent
 - <http://example.com:5000/probe>
 - <http://example.com:5000/current>
- Understand the implementation version:
 - Refer to `<Header version="..." />` and/or the XML namespaces in the root node
- Review each element:
 - Tag name
 - Position in the node tree
 - Attributes
- Compare against the MTConnect Standard and version:
 - See <https://model.mtconnect.org/>



Levels of Structural Non-Compliance

- **XML:** Violations of W3C specification for [XML v1.0](https://www.w3.org/XML/)
- **XSD:** Violations of W3C specification for [XML Schema v1.1](https://www.w3.org/TR/xmlschema11-1/)
- **MTConnect:** Violations of [MTConnect specification v1.0+](https://model.mtconnect.org/index.html)

<https://www.w3.org/XML/>
<https://www.w3.org/TR/xmlschema11-1/>
<https://model.mtconnect.org/index.html>

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Examples and Side Effects of Structural Non-compliance

Example

XML violation, malformed XML

```
<?xml version="1.0" encoding="UTF-8" ?>
<MTConnectDevices xmlns:xsi="..." xmlns="...">
  <Devices>
    <!-- Continued device model -->
    [REDACTED]
  </MTConnectDevices>
```

Fix

Add missing end tag

```
<?xml version="1.0" encoding="UTF-8" ?>
<MTConnectDevices xmlns:xsi="..." xmlns="...">
  <Devices>
    <!-- Continued device model -->
    </Devices>
  </MTConnectDevices>
```

Impact

- XML read errors

Examples and Side Effects of Structural Non-compliance

Example

XSD violation, non-unique id properties

```
<DataItem id="exec" name="path1execution" type="EXECUTION" />  
<DataItem id="exec" name="path2execution" type="EXECUTION" />
```

Fix

Change ids

```
<DataItem id="exec1" name="path1execution" type="EXECUTION" />  
<DataItem id="exec2" name="path2execution" type="EXECUTION" />
```

Impact

- Conflicting unique keys in database
- Misrepresented data

https://model.mtconnect.org/#Structure_EAID_002C94B7_1257_49be_8EAA_CE7FCD7AFF8A

Examples and Side Effects of Structural Non-compliance

Example

XSD violation, invalid datatype

```
<Samples>  
  <Load dataItemId="xload" name="XLoad" ... >TEST</Load>  
</Samples>
```

Impact

- String cannot be saved in a floating-point field
- Data loss
- Reporting inconsistency
- Poorly trained AI (garbage in, garbage out)

Fix

Output expected float result

```
<Samples>  
  <Load dataItemId="xload" name="XLoad" ... >70</Load>  
</Samples>
```

Examples and Side Effects of Structural Non-compliance

Example

MTConnect violation, duplicate names on the same level

```
<Device>
  <Components>
    <Axes id="laxes" name="Axes">
      <Components>
        <Linear id="lx1" name="X" />
        <Linear id="lx2" name="X" />
      </Components>
    </Axes>
  </Components>
</Device>
```

Impact

- Conflicting unique keys in database

Fix

Change names

```
<Device>
  <Components>
    <Axes id="laxes" name="Axes">
      <Components>
        <Linear id="lx1" name="X1" />
        <Linear id="lx2" name="X2" />
      </Components>
    </Axes>
  </Components>
</Device>
```

Examples and Side Effects of Structural Non-compliance

Example

MTConnect violation, invalid type

```
<DataItem category="SAMPLE" type="PATH_FEEDRATE_OVERRIDE" ... />
```

Impact

- Invalid type
- Unclear if intended type is
PATH_FEEDRATE_OVERRIDE or
PATH_FEEDRATE

Fix

Change category to EVENT

Change type to PATH_FEEDRATE

```
<DataItem category="EVENT" type="PATH_FEEDRATE_OVERRIDE" ... />
```

```
<DataItem category="SAMPLE" type="PATH_FEEDRATE" ... />
```

https://model.mtconnect.org/#Structure_EAID_B0781166_9F74_4844_B40C_CA32663868F2

https://model.mtconnect.org/#Structure_19_0_3_45f01b9_1580378218381_937176_2001

Examples and Side Effects of Structural Non-compliance

Example

MTConnect violation, invalid subtype

```
<DataItem category="SAMPLE" type="POSITION" subType="CURRENT" ... />
```

```
<DataItem category="SAMPLE" type="POSITION" subType="x:ACTUAL" ... />
```

Impact

- Conflicting unique keys in database
- Unclear if CURRENT is intended to be ACTUAL
- x:ACTUAL is an unnecessary extension

Fix

Change subtype to ACTUAL

```
<DataItem category="SAMPLE" type="POSITION" subType="ACTUAL" ... />
```

Examples and Side Effects of Structural Non-compliance

Example

MTConnect violation, missing units

```
<DataItem category="SAMPLE" id="x1load" name="X1Load" type="LOAD" />
```

Impact

- Unclear what units of LOAD may be
 - Newtons (N)?
 - Millivolts per Volt (mV/V)?
 - Percent (%)?

Fix

Change to PERCENT per standard

```
<DataItem category="SAMPLE" id="x1load" name="X1Load" type="LOAD" units="PERCENT" />
```

https://model.mtconnect.org/#Structure_EAID_B0781166_9F74_4844_B40C_CA32663868F2

https://model.mtconnect.org/#Structure_19_0_3_45f01b9_1580378218342_32796_1902

Examples and Side Effects of Structural Non-compliance

Example

MTConnect violation, invalid enumerable result

<Execution ... >6</Execution>

Impact

- Applications expect controlled vocabulary, exceptions thrown
- Failure to report [effectively]

Fix

Change to EXECUTION enumerable value

<Execution ... >PROGRAM_STOPPED</Execution>

Examples and Side Effects of Structural Non-compliance

Example

MTConnect violation, invalid axis name

```
<Device>
  <Components>
    <Axes id="laxes" name="Axes">
      <Components>
        <Linear id="lfoo1" name="foobar" />
      </Components>
    </Axes>
  </Components>
</Device>
```

Impact

- Applications expects ISO 841 axis names
- Unclear which axis from id

Fix

Change to ISO 841 name

```
<Device>
  <Components>
    <Axes id="laxes" name="Axes">
      <Components>
        <Linear id="lfoo1" name="X" />
      </Components>
    </Axes>
  </Components>
</Device>
```

How do you validate MTConnect Behavior?

- Test Agent REST endpoints
 - <http://example.com:5000/probe>
 - <http://example.com:5000/current>
 - <http://example.com:5000/assets>
- Test Agent REST endpoints with query parameters
 - <http://example.com:5000/sample?interval=1000>
- Compare against the MTConnect Standard and version:
 - See https://model.mtconnect.org/#Package_19_0_4_45f01b9_1637706615628_274120_5004

Levels of Behavioral Non-Compliance

- **HTTP:** Violations of RFC 7230 for [HTTP/1.1](#)
- **URI:** Violations of RFC 3986 for [URI](#)
- **MTConnect:** Violations of [MTConnect specification v1.0+](#)

<https://www.rfc-editor.org/info/rfc7230>

<https://www.rfc-editor.org/info/rfc3986>

https://model.mtconnect.org/index.html#Package_19_0_4_45f01b9_1637706615628_274120_5004

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Examples and Side Effects of Behavioral Non-compliance

Example

MTConnect violation, non-sequencing sequence number

```
<PathFeedrateOverride ... sequence="11">0</PathFeedrateOverride>  
<PathFeedrateOverride ... sequence="11">100</PathFeedrateOverride>  
<Load ... sequence="11">23</Load>
```

Impact

- Database key constraint violations
 - Composite keys could be generated from Header instance ID and observation sequence number.

Fix

Auto-increment sequence numbers

```
<PathFeedrateOverride ... sequence="11">0</PathFeedrateOverride>  
<PathFeedrateOverride ... sequence="12">100</PathFeedrateOverride>  
<Load ... sequence="13">23</Load>
```

Examples and Side Effects of Behavioral Non-compliance

Example

MTConnect violation, non-unique instance ID

```
<Header instanceId="1001" ...>
```

```
{{Restart Agent}}
```

```
<Header instanceId="1001" ...>
```

Impact

- Database key constraint violations
 - Composite keys could be generated from Header instance ID and observation sequence number.
- Duplicate data

Fix

Auto-increment instance ID upon startup

```
<Header instanceId="1001" ...>
```

```
{{Restart Agent}}
```

```
<Header instanceId="1002" ...>
```

Alternatively, instanceId may be the same if using a persistent buffer, but you must continue sequence numbers from last session

How do you validate MTConnect Capabilities?

- Define or identify requirements
- Compare requirements against Device model (e.g. Devices.xml or probe response)
- Validation Working Group developing recommendations for use case validation using W3C semantic web standards:
 - [OWL](https://www.w3.org/2001/sw/wiki/OWL)
 - [SHACL](https://www.w3.org/2001/sw/wiki/SHACL)

<https://www.w3.org/2001/sw/wiki/OWL>
<https://www.w3.org/2001/sw/wiki/SHACL>

Levels of Capability Non-Compliance

- **MTConnect:** Violations of [MTConnect specification v1.0+](#)
- **Use Cases:**
 - Applications (e.g. monitoring platforms, AI, ERP/MES)
 - Customers

<https://model.mtconnect.org/index.html>

Examples and Side Effects of Capability Non-compliance

Example

Monitoring Platform Requires:

- EXECUTION
- PROGRAM ACTIVE
- PATH_FEEDRATE_OVERRIDE
- ROTARY_VELOCITY_OVERRIDE

Impact

- Failed reporting
- Missing functionality

Examples and Side Effects of Capability Non-compliance

Example

Customer Requires:

- PROGRAM
- TEMPERATURE
- MATERIAL

Impact

- Non-conformance report
- Failed reporting
- Incomplete data

How do you handle non-compliance?

- Ask OEM/Machine Tool Builder for Native Solution
- Look for third-party adapters
- Look for third-party custom adapter developer

Why can't the computer just do it?

Tired of reading boring XML and pouring dozens of hours into validating an implementation? In this section we discover how Standards Committee members are developing automated solutions for validation.

Why can't the computer just do it?

Validation Working Group: Agent Active Self-Validation

- v2.5+ supports validation, quality, deprecated, and extended properties
- Covers some structural adherence
- Implementations that support active self-validation in the Agent contain the following in Streams response:

<Header validation="true" ... />

- Violations to vocabulary will be indicated by:

<Execution quality="INVALID" ...>6</Execution>

- Violations reported in MTConnectExceptionsReport Response
- Self-validation implementations can be supported by Transpiler

Why can't the computer just do it?

Validation Working Group: MTConnect Transpiler

- MTConnect Transpiler is a SDK to simplify MTConnect-related software development/maintenance
- Validation Working Group supports open-source development
- [MtconnectTranspiler](https://github.com/mtconnect/MtconnectTranspiler) deserializes MTConnect Standard from SysML to C#
- [MtconnectTranspiler.Sinks.CSharp](https://github.com/mtconnect/MtconnectTranspiler.Sinks.CSharp) is the MTConnect Standard represented in navigable C#
- MTConnect Standard updates done in minutes using Transpiler

<https://github.com/mtconnect/MtconnectTranspiler>

<https://github.com/mtconnect/MtconnectTranspiler.Sinks.CSharp>

Why can't the computer just do it?

Validation Working Group: SHACL Validation

- Validation Working Group developing recommendations for using [SHACL](#)
- Translate MTConnect to RDF and then validate shapes

<https://github.com/mtconnect/MTConnectShacl>

```
val: LinearAxesMustBeXYOrZ
  a sh:NodeShape ;
  sh:targetClass mt:Linear ;
  sh:property [
    sh:path mt:hasName ;
    sh:pattern "[X-Z][0-9]*$";
    sh:message "name must be one of X, Y, or Z followed by an
optional number" ;
    sh:minCount 1 ;
  ] .
```

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Can also use SPARQL for complex validation

```
val:IdsMustBeUnique
  a sh:PredicateShape ;
  sh:targetSubjectsOf mt:hasId ;
  sh:sparql [
    sh:prefixes val:Prefix ;
    sh:message "All `id` properties **MUST** be unique." ;
    sh:select """
      SELECT $this ?id WHERE {
        $this mt:hasId ?id .
        ?other mt:hasId ?id .
        FILTER ($this != ?other)
      }
      """ ;
  ] .
```

Why can't the computer just do it?

Validation Working Group: Commercial Options

- Metalogi
 - Active self-validating MTConnect Agent
- True Analytics Manufacturing Solutions (TAMS)
 - <https://validator.tams.ai/>
 - Response Document validation covers most structural adherence
 - Basic use case validation
 - Agent Auditor
 - Active validation of one or more Agent(s)

Exception Reporting

- Added in v2.6
- <https://example.com:5000/errors>
- Provides a list of validation exceptions (e.g. quality="INVALID")

<Exception

severity="ERROR|FATAL|INFO|WARNING"

code="DEPRECATED|DUPLICATE_ENTRY|EXTENDED|INVALID_FORMAT|NOT_FOUND|OUT_OF_RANGE|TYPE_MISMATCH">

<Message><!-- Exception message --></Message>

<Trace>

<Source><!-- URI of the artifact of where the exception occurred. --></Source>

<Context><!-- unique path to the exception location. --></Context>

<ScopeType><!-- designates the exception property kind. --></ScopeType>

<Scope><!-- name of the property. --></Scope>

<Value><!-- given value of the property. --></Value>

<ExpectedType><!-- normative value type for the property. --></ExpectedType>

</Trace>

</Exception>

Tools for Validation

- Notepad++
 - XML Tools Plugin
- Visual Studio Code
- Microsoft Excel
- Reference C++ Agent (Active Validation) (<https://github.com/mtconnect/cppagent>)
- Metalogi MetaAgent (<https://www.metalogi.io/products/metaagent>)
- TAMS Online Validator + Agent Auditor (<https://validator.tams.ai/>)

Lab: Validating by Hand

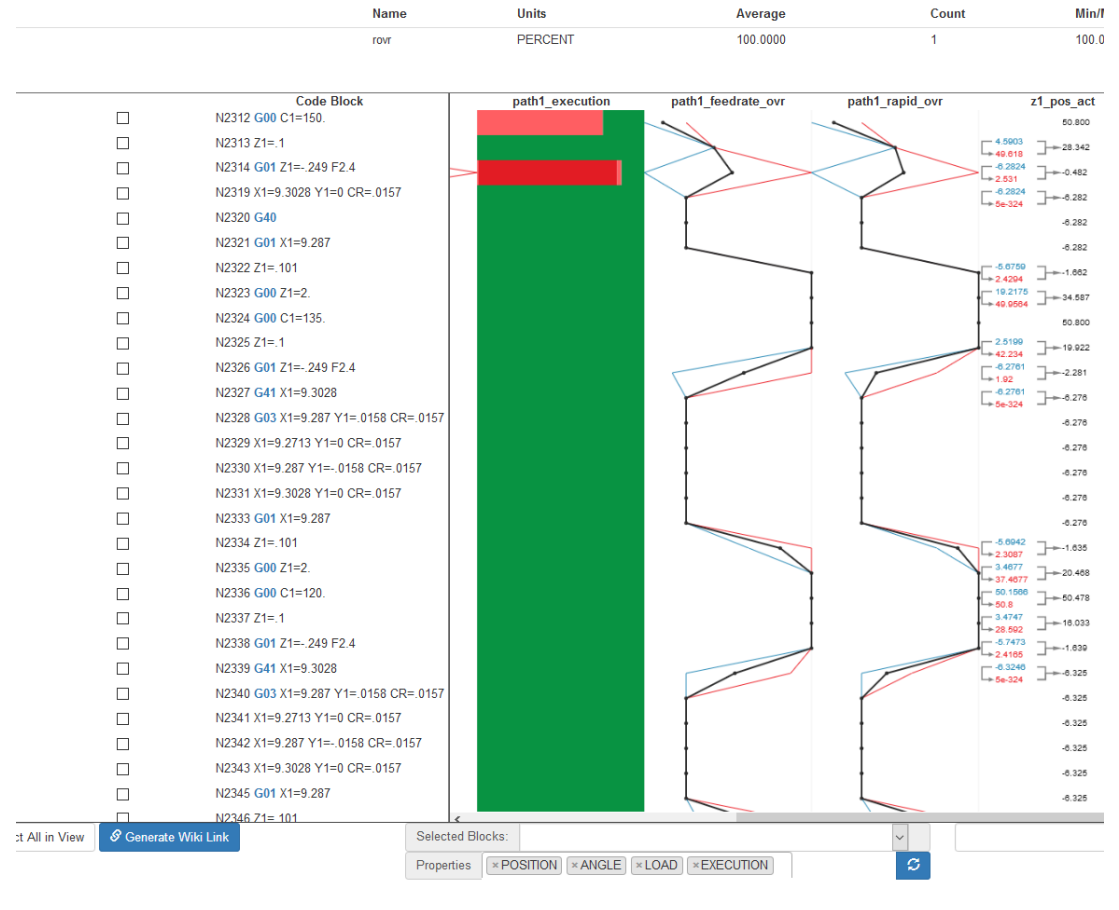
- Use Notepad++ to review XML
- Use <https://model.mtconnect.org> as reference

Lab: Validating with TAMS Online Validator

- Upload XML to <https://validator.tams.ai>
- Review + navigate report

Use-Cases and Demos

- Story Time: “I don’t know what happened...”
([LinkedIn Article](#))
- TAMS Online Validator (<https://validator.tams.ai/>)
 - Free spot-checks on MTConnect XML responses
 - Validation exception reports
 - Use-Case validation
- TAMS Shop-Connect
 - Avoid mapping tags with plug-and-play dashboards
- Chris Misztur, Mr. IIoT – LLM, Dashboards, Open-Source (<https://www.linkedin.com/in/chrismisztur/>)
- Eunseob Kim, Purdue – LLM, high frequency data
(<https://www.sciencedirect.com/science/article/pii/S2213846324002499?via%3Dihub>)



Open-Source Libraries

- Transpiler (<https://github.com/mtconnect/MtconnectTranspiler>)
 - C# (<https://github.com/mtconnect/MtconnectTranspiler.Sinks.CSharp>)
 - Python (<https://github.com/mtconnect/MtconnectTranspiler.Sinks.Python>)
 - C++ (<https://github.com/mtconnect/MtconnectTranspiler.Sinks.CPlusPlus>)
 - JSON (<https://github.com/mtconnect/MtconnectTranspiler.Sinks.JsonSchema>)
- W3C
 - SHACL+SPARQL (<https://github.com/mtconnect/MTConnectShacl>)
 - OWL
 - Manchester
- Client Apps
 - Ladder99 (<https://github.com/Ladder99/ladder99>)
 - TrakHound (<https://github.com/TrakHound/TrakHound>)

Need more advanced help?

- Vendors that support validation:
 - Metalogi
 - TAMS
- Attend Standards Committee meeting(s) to stay up-to-date with vendor lists

Links to Documents

<https://www.mtconnect.org/>

<https://www.mtconnect.org/documentation>

<https://model.mtconnect.org/>

<https://github.com/mtconnect/cppagent>

MTConnect Institute

- <https://www.mtconnect.org>
- <https://www.mtconnect.org/getting-started>
- <https://www.mtconnect.org/documentation>
- <https://model.mtconnect.org>
- <https://projects.mtconnect.org> (requires registration & login)
- The MTConnect Institute has a number of **Working Groups** dedicated to the Standard's development and promotion.
- E-Mail: bootcamp@projects.mtconnect.org

Q&A